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Liquefied natural gas trade network changes and its mechanism in the context of the Russia–Ukraine conflict

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ABSTRACT

Liquefied natural gas (LNG), as a transitional fossil fuel, plays a vital role in the modern energy transition process. In the context of the Russia–Ukraine geopolitical conflict, studying the evolution pattern and mechanisms of LNG trade networks is crucial for maintaining global energy security, particularly for countries relying on LNG imports. The study is based on multisource big data from 2021 to 2022, including Automatic Identification System (AIS) data, Gdelt news data, and remote sensing satellite data. It utilizes complex network metrics and the temporal exponential random graph model (TERGM) method to analyze the evolution patterns and mechanisms of the global LNG trade network. The research findings indicate the following: (1) The Russia–Ukraine conflict has led to a significant increase in LNG imports by European countries. Many countries have opened temporary direct shipping routes, which has accelerated the decline in the average path length of the network and improved network efficiency. (2) The global LNG trade community trend has been strengthened by the Russia–Ukraine conflict. The number of members within the trading community in which the United States is located has increased, with some European countries moving into this community. (3) The reciprocal structural characteristics of the LNG trade network drive its development, and the expansion of the national port handling capacity ability also drives network growth. Conversely, large differences in geopolitical relations, culture, and levels of governance between countries can hinder global LNG network development. This study provides a theoretical basis and decision-making reference for energy security under major geopolitical conflicts.

1. Introduction

Natural gas, as a transitional fossil fuel toward renewable clean energy, is increasingly asserting its strategic importance. In recent decades, the burning of fossil fuels has led to a significant increase in carbon dioxide emissions, thus accelerating global climate change. Global warming has led to serious consequences, such as rising sea levels, increased frequency of extreme weather events, reduced biodiversity, and decreased crop yields (Shivanna, 2022). Compared with coal, natural gas-fired generation emits about 30 to 50 % less carbon, thereby making it a critical bridge for achieving a large-scale low-carbon transformation by replacing coal power (Myhrvold and Caldeira, 2012). Currently, global demand for natural gas is rapidly increasing. BP company forecasts that LNG will surpass coal to become the second-largest energy source by 2040. In the future, the role of LNG in the green and low-carbon transition process will continue to strengthen.

Geopolitical conflicts bring significant uncertainty to the security of LNG supply. In February 2022, the Russia–Ukraine conflict erupted, which led to a significant use of LNG from the United States to fill the European market (Lambert et al., 2022). Compared with overland pipeline transportation, LNG transportation has the advantage of being able to traverse oceans, thus offering a flexible supply mode that reduces the constraints of traditional supply chains. Compared with onshore pipeline transportation, the advantage of LNG transportation across seas offers a flexible supply mode, which mitigates spatial constraints associated with the pipeline supply. In the face of various challenges and

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uncertainties in the LNG supply chain, a thorough analysis of the impact of geopolitical relations on LNG trade networks is essential. This examination not only helps relevant countries establish LNG supply guarantee mechanisms in response to geopolitical risks but also contributes to the theoretical improvement of LNG trade network analysis.

Scholars (Peng et al., 2019; Xiao et al., 2024) have conducted extensive research on the topological characteristics and evolution patterns of the global LNG trade network. However, relatively limited discussions have been initiated on the formation mechanism of the trade network (Guo et al., 2023a). Existing research (Gupta et al., 2019) primarily focuses on the impact of geopolitical relations on bilateral or regional trade, with results indicating a complex mechanism of geopolitical influence. For instance, accidental diversion theory suggests that deteriorating geopolitical relations may inhibit bilateral trade, while the dependence hypothesis posits that import-dependent countries are unlikely to alter the status quo, with deteriorating geopolitical relations having no significant negative impact on trade (Agarwal and Golley, 2022). Given the unique characteristics of multilaterality, heterogeneity, and dynamism in LNG networks (Guo et al., 2023a; Roudari et al., 2023), the mechanisms through which geopolitical relations affect the LNG trade network may differ from those in bilateral trade. Therefore, the influence of various factors, including geopolitical relations, on the topological structure and dynamic evolution of the LNG trade network must be comprehensively examined.

The main contributions of this research are primarily manifested in two aspects. (1) Utilizing AIS data, we construct two annual trade networks in 2021 and 2022 to directly analyze the changes in trade patterns. Global LNG trade statistics are often challenging to obtain in a timely manner. For example, Qatar's LNG trade data in the UN database is only updated through 2019. AIS overcomes this issue by facilitating timely tracking of LNG trade. It provides real-time data on a vessel's location, course, speed, destination, and other relevant information through a radio communication system (Xiao et al., 2024a). (2) Based on the Temporal Exponential Random Graph model (TERGM), this study integrates high-frequency, multi-source big data to explore the mechanism of LNG trade network. The TERGM model is effective for analyzing network evolution as it captures both stability and changes in network structures over time (Wang et al., 2024a). Long term data is essential for the Tergm model. While this paper constructs 24 monthly LNG trade networks from 2021 to 2022, capturing monthly exogenous factors like geopolitical relations remains a significant challenge. To align with the monthly trade networks, this study utilizes multisource big data to evaluate influential indicators, including GDELT for geopolitics, nightlights for economic activity, and Sentinel NO2 for environmental shifts.

The remaining sections of this study are organized as follows. Section 2 provides a review of existing literature. Section 3 introduces the methods and data used. Section 4 presents the patterns and mechanisms of the LNG trade network. Sections 5 and 6 discuss the findings and conclude the paper, respectively.

2. Literature review

2.1. Research on liquefied natural gas trade

LNG trade is crucial to global energy security and has been analyzed from various perspectives within the academic community. Scholars investigate the spatiotemporal differentiation of LNG trade. For instance, Zhu et al. (2023) analyzed the spatiotemporal evolution of LNG competition patterns, while Yan et al. (2023) examined the spatiotemporal changes in China LNG import voyages. Complex networks can more accurately represent the dependencies between nodes and are widely used in international trade (Wang et al., 2024a). Scholars have analyzed LNG trade from the perspective of complex networks. Geng et al. (2014) analyzed the overall evolution of the global LNG network structure using indicators such as network density, average clustering coefficient, and average path length. Peng et al. (2021) have focused on analyzing the characteristics of local community structures within networks using indices such as modularity.

In studying the formation mechanism of LNG trade, scholars have paid relatively scarce attention to the time effect of the LNG trade network. Although early classical gravity models have been widely used for bilateral trade flow analysis, they have fallen short in capturing self-organizing network patterns (Wang et al., 2024a). Recently, the exponential random graph model (ERGM) has seen increased use as a probabilistic generative model that accounts for endogenous network structures and exogenous attribute effects (Liu et al., 2022). The ERGM, which utilizes maximum likelihood estimation, offers a comprehensive assessment of structural impacts on network formation (Robins et al., 2007). Furthermore, ERGM's superior parameter interpretability and model construction flexibility make it a more advanced approach than the quadratic assignment procedure (Pan et al., 2022). However, ERGM uses cross-sectional network data for the same time point (Broekel et al., 2014), this model is still static. To capture the temporal change features of the network, ERGM can be extended to incorporate previously observed lagged networks, which is called TERGM. TERGM models multiple networks over time as a whole, simultaneously examining endogenous structural and temporal dependencies, and is considered a higher-order model compared to ERGM (Pan et al., 2022).

2.2. Analysis of factors influencing the LNG trade network

The dynamics of global trade networks are intricately complex. The existing literature (Guo et al., 2023b; Wang et al., 2024a) highlights the pivotal roles of geopolitical ties, logistical efficiency, economic disparities, and geographic proximity. These factors are broadly classified into endogenous structural, exogenous national attributes, and external network embedding effects.

2.2.1. Endogenous structural effects

Endogenous structural effects are key drivers in shaping trade relationships (Wang et al., 2024a; Wang et al., 2024b). Specifically, reciprocity, which indicates mutual links between network nodes, promotes information exchange and reduces informational asymmetry (Garlaschelli and Loffredo, 2004); Connectivity (Gwdsp), a geometrically weighted metric, measures the spatial proximity and likelihood of trade connections between countries via intermediaries (Pan et al., 2022); Transitivity (Gwesp) measures the extent of triadic closure in networks. High levels indicate increased trust and cooperation among nodes, which subsequently facilitates the establishment of trade relationships. Activity (Gwodeg) measures the attractiveness of nodes for new connections, with high activity levels increasing the likelihood of wealthy high-degree nodes accumulating many connections.

2.2.2. Exogenous national attribute effects

Exogenous national attributes, such as economic level, have a significant impact on the formation and maintenance of trade networks (Wang et al., 2024a; Wang et al., 2024b). These node attributes can reflect the economic attractiveness and accessibility between countries, thereby affecting the probability of connections between nodes in the trade network (Yan et al., 2024).

Trade logistics is a important factor in facilitating international production networks (Saslavsky and Shepherd, 2014). Approximately 80 % of global trade by volume is conducted through maritime transport (UNCTAD, 2022). As a fundamental link in international trade, the efficiency of port logistics plays a crucial role in facilitating global trade. Efficient port operations can markedly cut LNG logistics transportation costs (Clark et al., 2004). Port logistics efficiency is often defined by statistical indicators such as maximum ship size, ship calls, and number of berths (Feng et al., 2020; Zhao et al., 2017). Recently, scholars have utilized AIS data to redefine port efficiency through metrics such as vessel docking time, anchorage duration, and cargo unloading time at ports (Feng et al., 2020; Zhong et al., 2020). AIS provides real-time vessel tracking through satellite and coastal station data exchange (Xiao et al., 2024), thus offering a precise measure of port logistics efficiency at the national level.

A country's environmental pollution and economic development levels are key exogenous node attributes influencing trade networks. The environmental Kuznets curve (EKC) posits that pollution increases with industrialization during early economic growth but declines as countries with high GDP per capita pursue clean energy (Dinda, 2004). Researchers have validated the theories mentioned above regarding the relationship among environmental pollution, economic development level, and trade levels (Sheehan et al., 2014). Pollutant concentrations such as $PM_{2.5}$, CO_2, and NO_2 are used to measure environmental pollution (Dechezleprêtre and Sato, 2017), while night light is used as a proxy for GDP (Li et al., 2019). However, in the context of the Russia–Ukraine conflict, EKC theory is not applicable to some countries. Developed countries, such as Germany, Finland, and Italy, have restarted coal-fired power plants to lessen reliance on Russia, which has affected progress toward their environmental targets (Kuzemko et al., 2022; Huang et al., 2023). Hence, the intrinsic linkages among LNG trade networks, economic development levels, and environmental pressures on a global scale must be further explored.

2.2.3. External network embedding effects

Academic research has examined embedded external networks, such as geographic distance, language, and regional trade agreements (Wang et al., 2024a; Guo et al., 2023b), with a notable recent increase in attention to geopolitical networks. Deteriorating geopolitical relations can negatively impact trade via trade barriers, sanctions, and infrastructure damage (Khan et al., 2023). However, geopolitical relations involve a complex set of military, economic, cultural, and ideological factors that interact and change dynamically.

Quantifying geopolitical relations remains an academic challenge. Many scholars characterize geopolitical relations using United Nations voting data (Bailey et al., 2017). Additionally, some think tanks, such as The Economist Intelligence Unit, The Political Risk Services Group, and Business Monitor International, calculate geopolitical relations by selecting a set of relevant variables and weighing them (Xiong et al., 2020). However, these methods are often biased by subjective factors in the selection and weighting of variables or limited by long statistical periods that disguise the dynamics of high-frequency political events.

The Gdelt project has addressed these limitations by capturing geopolitical events from global news with a 15-min temporal resolution across more than 150 countries (Alipour et al., 2024). Moreover, Gdelt's Goldstein scores are empirically derived and based on big data standards. These standards clarify the scoring range for different events, thus reducing subjective weighting space and enhancing the comparability and authority of the results.

2.3. Limitations of the existing literature

After the outbreak of the Russia-Ukraine conflict, scholars followed the previous research perspective to conduct in-depth research on LNG trade. For example, scholars (Xiao et al., 2024) have focused on the resilience changes of LNG trade networks from a network perspective, while others (Zhang et al., 2024) have analyzed the spatiotemporal evolution of LNG supply patterns based on route trajectories. Although the academic community has paid some attention to the changes in LNG trade after this conflict, there is relatively little research on the LNG trade network driving mechanisms. This gap may be due to several limitations:

The construction of LNG trade network requires support from higher frequency data. Many studies on LNG trade networks (e.g., Guo et al., 2023b; Zhu et al., 2023) primarily rely on annual statistical data. However, world trade statistics are often characterized by delays and inconsistencies (Parniczky, 1980), which hinder the timely tracking of LNG trade network changes. In addition, geopolitical events, known for their suddenness, require a more timely and consistent dataset to accurately assess their impact on the LNG network. Relying solely on annually published trade statistics may result in an underestimation or delayed response to these events.

A more comprehensive framework is required for the analysis of the influence mechanisms within LNG trade networks. Scholars have explored the mechanisms affecting trade networks from various factors such as the environment, logistics efficiency, geopolitical relationships, and the economy (Wang et al., 2024a; Wang et al., 2024b). However, existing research seldom places these variables within a unified framework to discuss the evolution mechanisms of LNG trade networks. This may be attributed to various factors. On one hand, LNG trade encompasses multiple disciplines, including economics, logistics, and geopolitics, where the selection or construction of influencing variables necessitates support from disciplinary knowledge. For instance, the efficacy of indicators such as the maximum vessel size in measuring logistics efficiency still requires further validation. On the other hand, although high-frequency monthly trade data helps to reveal the dynamic evolution mechanism of networks. This also necessitates that the relevant variables have the same temporal resolution, which in turn increases the complexity of integrating different analyses.

To address these shortcomings, this study utilizes AIS data from 2021 and 2022 to construct two trade networks, providing a basis for directly comparing changes in the trade network under the impact of the conflict. Second, we used multi-source big data to quantify impact factors on a monthly scale. This paper constructs more precise indicators of port logistics efficiency using AIS data. Port logistics efficiency is defined as the time spent by port handling units loading and unloading LNG per unit of weight. Furthermore, this study employs big data to quantify additional influencing factors, including GDELT for geopolitical analysis, Black Marble night light data for economic activity, and Sentinel NO_2 data for environmental changes. To enhance the efficiency of large-scale data processing, this study employs BigQuery to process Gdelt data and Google Earth Engine to handle remote sensing data. Finally, this study utilizes the TERGM model to assess the impact of various factors on the network, including endogenous structural elements, exogenous national attributes, and external network embedding effects (Fig. 1).

3. Methods and data

3.1. Data preprocessing

3.1.1. LNG trade network construction

The construction of the LNG trade network is the cornerstone of this study. The AIS trajectory data of LNG carriers from 2021 to 2022 are extracted based on the Maritime Mobile Service Identity numbers. This process yielded a total of 246 million trajectory records from 580 LNG carriers. Voyages are identified by integrating AIS data (port calls) with port vector map information (loading or unloading ports). The AIS data was sourced from Shipxy (https://www.shipxy.com/) and GOGOTRADE (http://www.boloomo.com/product/5.cshtml), both recognized reputable and reliable data providers based in Beijing, China.

In this study, we constructed undirected weighted networks $G = (V, E, W)$ using LNG trade data from 2021 to 2022. We abstracted 57 countries (or regions) as network nodes,denoted as $V = \{v_1, v_2, \dots, v_{57}\}$. The trade relations between countries (or regions) are represented by ordered pairs (i, j), where $vi, vj \in V$, $i \neq j$. $E = \{E_i\}$ represents the set of all edges within V. if there is LNG trade between countries i and j, then $E_{ij} = 1$, otherwise, $E_{ij} = 0$. $W = \{w_{ij}\}$ denotes the weight matrix for the network edges, with the weight between nodes i and j determined by the sum of deadweight tonnage of vessels traded between countries.

In terms of pattern changes, this paper constructs two LNG trade networks for 2021 and 2022 to directly examine the trade shifts under the impact of the Russia-Ukraine conflict. This is because maritime shipping typically takes several months to complete, and annual data better captures the cumulative changes occurring throughout the shipping process. In terms of impact mechanisms, this paper constructs and

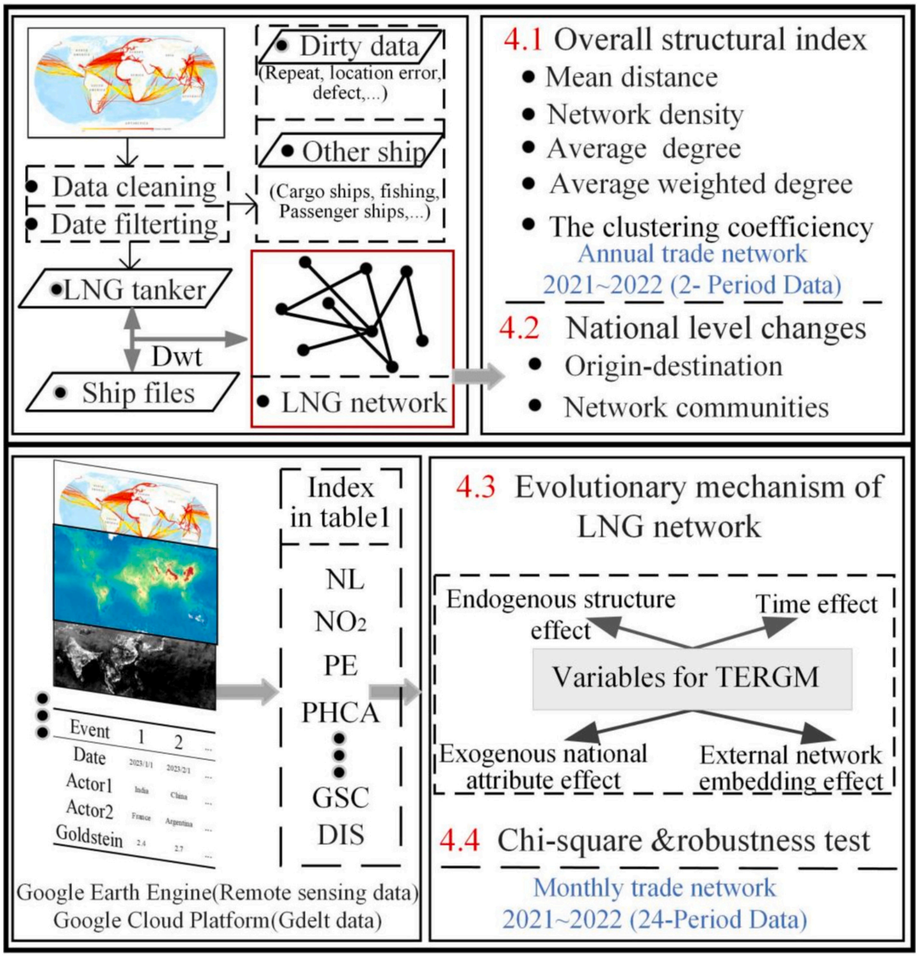


Fig. 1. Research framework.

analyzes 24 distinct LNG trade networks based on monthly data from 2021 to 2022. This is because factors such as geopolitical relations are highly time-sensitive, and their impact on trade often has a lag effect. Monthly data allows for a more detailed examination of the dynamic changes in the LNG trade network mechanism and the network's temporal dependency effects.

3.1.2. Calculation of geopolitical relations

The study focused on examining the bidirectional nature and dynamic characteristics of geopolitical relations. The Gdelt database utilizes the Goldstein conflict-cooperation score system (Goldstein, 1992) to encode and score over 300 types of reported events. This scoring system effectively quantifies the degree of conflict or cooperation in reported events through the semantic analysis of the language used in event reports. However, many indices of geopolitical relations based on the Gdelt suffer from several limitations. For example, Huang et al. (2023) focused primarily on the unidirectional average scores of a single country's relationship with others; however, they overlooked the reciprocal nature of geopolitical relationships. Similarly, Guo et al. (2023a) used aggregated Goldstein scores on an annual basis, but the time resolution was still rough. To address these shortcomings, the following improvements are proposed in this study.

$$\mathrm{DayGS}_{a\to b} = GS_1 \times \frac{C_1}{Call} + GS_2 \times \frac{C_2}{Call} + \ldots GS_n \times \frac{C_n}{Call} \tag{1}$$

$$\mathrm{MGS}a - b = \frac{\sum \mathrm{DayGS}_{a\to b} + \mathrm{DayGS}_{b\to a}}{Cday} \tag{2}$$

where $\mathrm{DayGS}_{a\to b}$ represents the weighted sum of the Goldstein scale values of country A toward country B on a specific day. GS_n represents the Goldstein scale value of the nth event, and C_{all} represents the total number of events on that day. $\mathrm{MGS}_{a\text{-}b}$ represents the weighted monthly Goldstein scale value between countries A and B, and C_{day} represents the number of days in the month with Goldstein scale scores. Finally, we constructed undirected Geo Relational Networks (GRN) using the MGSa-b values between countries as weights.

3.1.3. Processing of other big data

The NO_2 concentrations of various countries are utilized to gauge the environmental pollution levels, with the NO_2 data sourced from the TROPOMI product. The nighttime light intensity of various countries serves as an indicator of economic development levels, with the nighttime light data derived from the Black Marble product. Port logistics efficiency is defined as the time spent by port handling units loading and unloading LNG per unit of weight. The port logistics efficiency index is calculated based on AIS data. Appendix A presents the detailed processing methods of the aforementioned indicators.

3.2. Research methods

3.2.1. Network community analysis

Network communities classify relatively tightly connected clusters of nodes that reflect subnetworks or local regions of the network in terms of spatial and functional dimensions (Pu et al., 2019). Connections within the same community consider the relationship between two nodes and the relationship between nodes associated with the target node (Wang et al., 2022).

The global LNG trade exhibits regional characteristics influenced by factors such as transportation costs, resource availability, and infrastructure (Barnes and Bosworth, 2015; Chen et al., 2019). However, geopolitical conflicts with trade restrictions and rising energy costs can disrupt regional trade and change trade communities (Cui et al., 2023). We examine the network communities using the modularity index, with the specific formula as follows:

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{i,j} - \frac{k_i k_j}{2m}\right]\delta(C_i, C_j) \tag{3}$$

A_{ij} represents node i, the weight between j; K_i represents node strength. C_i represents the community to which node i belongs; When $C_i = C_j$, δ (C_i, C_j) = 1; otherwise, $\delta(C_i, C_j) = 0$. m represents the sum of the weights of all edges.

The algorithm introduced by Blondel et al. (2008) was utilized to partition the LNG network into communities, each symbolizing a cluster of countries. This algorithm operates in two distinct phases. Initially, each node is considered an individual community. The change in modularity for each node i and its neighboring community C is expressed as ΔQ. node i joins the neighboring community with the highest positive ΔQ; if no positive ΔQ exists, it remains in its original community.

$$\Delta Q = \left[\frac{\sum C^{in} + 2k_i^{in}}{2m} - \left(\frac{\sum C^{tot} + k_i}{2m}\right)^2\right] - \left[\frac{\sum C^{in}}{2m} - \left(\frac{\sum C^{tot}}{2m}\right)^2 - \left(\frac{\sum k_i}{2m}\right)^2\right] \tag{4}$$

Where ΣC^{in} is the sum of the weights of the edges inside the neighboring community C. ΣC^{tot} is the sum of the weights of the edges incident to all nodes in C, k_i represents the sum of the weights of the edges of node i, k_i^{in} is the sum of the weights of edges from node i to nodes in community C, and m is the total edge weight of the network. An analogous formula assesses the modularity shift when node i is extracted from its current community. Practically, this involves calculating the modularity difference by first removing i from its community and subsequently transferring it to an adjacent community.

In the second phase of the algorithm, a new network structure is formed where nodes are the communities identified in the first phase. Edges between nodes within the same community result in self-loops in the new network. Once the new network is constructed, the algorithm's first phase is reapplied, refining the community structure iteratively. The process repeats until the community count stabilizes and maximum modularity is achieved, indicating optimal partitioning. As the calculation of overall network metrics is relatively common, this paper follows the approach of Geng et al. (2014) without duplicating the calculation process.

3.2.2. TERGM model

Most LNG trade involves the establishment of long-term contracts, and the interdependencies between historical and current network structures cannot be overlooked (Guo et al., 2023b). Additionally, the maritime transportation of LNG typically spans several months, suggesting that geopolitical conflicts may exert a delayed impact on LNG trade. Consequently, this study employs the TERGM model, which accounts for temporal dynamics. The TERGM framework encompasses endogenous structural effects, exogenous national attribute effects, and the effects of external network embeddings. The calculation equation is as follows:

$$P(N^t|\beta^t, N^{t-1}) = \left(\frac{1}{\alpha}\right) exp\begin{pmatrix} \beta_0 Edges + \beta_1 Mutual + \beta_2 Gwdeg + \beta_3 Gwesp + \beta_4 Gwdsp + \beta_5 Stability \\ +\beta_6 Variability + \beta_{i1}\ln NO_2 + \beta_{i2}\ln NL + \beta_{i3}\ln PE + \beta_{i4}\ln PHCA \\ +\beta_{j1}\ln NO_2 + \beta_{j2}\ln NL + \beta_{j3}\ln PE + \beta_{j4}\ln PHCA + \beta_{ij}GSC \\ +\beta_{ij}DIS + \beta_{ij}CON + \beta_{ij}FTA + \beta_{ij}CUL + \beta_{ij}GOV \end{pmatrix} \tag{4}$$

where N^t and N^{t-1} represent the LNG product trade networks during period t and period t-1, respectively. The 24 months from 2021 to 2022 constitute the 24 periods of this study. β is the corresponding estimated parameter, and $1/\alpha$ is the normalized parameter. *Edges* are edge variables in the trade relationship network. i and j represent the sender and receiver (which can be understood as the exporting and importing countries), respectively. The remaining variables can be found in Table 1. This article uses the Maximum Pseudolikelihood Estimation method to estimate the tergm model.

4. Changes in the LNG trade network

4.1. Overall changes in global LNG trade volume

The outbreak of the Russia–Ukraine conflict has changed the global LNG trade landscape. Energy independence from Russia has led to a rapid increase in LNG demand in Europe (Fig. 2a). The United States, with its coastlines on the Atlantic and Pacific, leverages its strategic geography to supply LNG to Europe and Asia. However, The United States redirected a substantial export portion of its LNG to Europe to compensate for the Russian gas deficit, thus resulting in a reduction in shipments to Asia in 2022 (Fig. 2b).

The United States allies in the Asia-Pacific are tightening LNG trade ties, with a concurrent trend of supply decline in neighboring areas. In the Asia-Pacific, The United States has forged alliances such as QUAD and IPEF to counterbalance China, with discussions on potentially integrating Japan and South Korea into NATO's framework (Koga, 2023). After the outbreak of the Russia–Ukraine conflict, imports of LNG from Australia to destinations such as Japan, South Korea, and Taiwan (China) have all increased (Fig. 2b). Conversely, most United States allies in the Asia-Pacific, especially Japan, which is involved in territorial issues with Russia, have experienced a decrease in LNG trade with Russia.

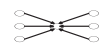
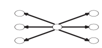
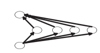
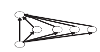
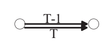
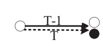
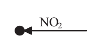
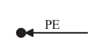
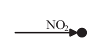
4.2. Overall network structural changes in the LNG trade network

From 2021 to 2022, the average weighted degree of the LNG trade network increased, thus indicating a rise in the average volume of LNG traded at the national level. The conflict between Russia and Ukraine has triggered an energy supply crisis in Europe, thereby prompting many European countries to increase imports of LNG to reduce their dependence on Russia (Fig. 2a). Apart from traditional consumers such as Japan, some emerging markets, including India and Pakistan, are also experiencing rapid demand growth in 2022, thereby contributing to the overall increase in global LNG trade.

The decline in average path length from 2021 to 2022 signifies a trend toward shorter distances between nodes, thus reflecting an increase in network efficiency. After the outbreak of this conflict, a significant natural gas deficit emerged in the European market, which has led to a surge in activity in the European LNG shipping market. Some European countries had to open temporary direct shipping routes to cope with the energy shortage caused by the "de-Russia" situation. In general, the reduction in transit segments within LNG transportation directly causes a decrease in the average path length of the LNG trade network.

From 2021 to 2022, the LNG trade network exhibited an increase in density, average degree, and clustering coefficient (Table 2), thus indicating improved network connectivity and enhanced local clustering. These shifts are intricately tied to the Russia–Ukraine conflict. The EU has levied a series of 10 sanctions against Russia, including restrictions on oil imports (Chen et al., 2023). However, the EU has refrained from imposing direct sanctions on Russian LNG, which are attributable to the significant dependency of Europe on Russian gas compared with oil. The 2022 sabotage of the Nord Stream 2 pipeline intensified Europe's energy crisis, thus prompting numerous countries to diversify their LNG imports to hedge against supply risks. Although these metrics of the LNG trade network have improved, this does not directly indicate that the network's supply security has been optimized. In fact, an increase in network density or clustering coefficients may conceal underlying risks within the network, particularly when the network is overly reliant on certain key nodes or pathways, potentially increasing the system's

Table 1
List of explanatory variables of TERGM.

Classification	Variable Name	Meaning	Structure	Data Source
Endogenous Structural Statistics	Edges	It serves as a basis for the formation of international LNG trade network relationships, being a constant factor.		—
	Mutual	The reciprocity tendency among nodes in the LNG trade network.		–
	Gwidegree	Checks the tendency of node convergence.		–
	Gwodegree	Checks the tendency of the node to expand.		–
	Gwdsp	The tendency for trading parties to establish LNG trade relationships indirectly through intermediaries.		–
	Gwesp	The trend of LNG trade bodies forming closed triangular trade relationships.		–
Time effect	Stability	The tendency for node relationships in the current trade network to persist and remain stable in the later stages of the network.		–
	Variability	The tendency for node relationships within the current trade network to be disrupted, while new node relationships are formed in the later stages of the network.		–
Exogenous national attribute effects	Receiver (NO_2)	The monthly average NO_2 concentration values for receiver.		Sentinel-5P data, Europe Space Agency
	Receiver (NL)	The monthly average nighttime light values for receiver.		Black Marble data, NASA
	Receiver (PLE)	Port logistics efficiency for receiver.		AIS data, Shipxy.com and GOGO-TRADE
	Receiver (PHCA)	Port handling capacity ability for receiver.		
	Sender (NO_2)	The monthly average NO_2 concentration values for sender.		Sentinel-5P data, Europe Space Agency
	Sender (NL)	The monthly average nighttime light values for sender		Black Marble data, NASA
	Sender (PLE)	Port logistics efficiency for sender.		AIS data, Shipxy.com and GOGO-TRADE
	Sender PHCA)	Port handling capacity ability for receiver.		
External network embedding effects	Edgecov (GRN)	The embedding effect of geopolitical relations network.		Goldstein data, Gdelt
	Edgecov (DIS)	Space distance embedding effect. The distance between the two countries' capitals is the numerical value of the network edge.		Geographical distance, CEPII
	Edgecov (CON)	Common geographic boundary embedding effect. If two countries share a border, the network edge value is 1; otherwise, it is 0.		Geographical distance, CEPII
	Edgecov (FTA)	FTA embedding effect. If both parties sign a trade agreement, the network edge value is 1; otherwise, it is 0.		Trade facilitation measures data, CEPII
	Edgecov (CUL)	Culture embedding effect. If two countries have similar cultures, the value of the network edge is 1; otherwise, it is 0.		Proxies for cultural proximity data, CEPII
	Edgecov (GOV)	Governance embedding effect. The difference in governance indices between two countries as the value of the edge.		Worldwide governance Indicators , World Bank Group

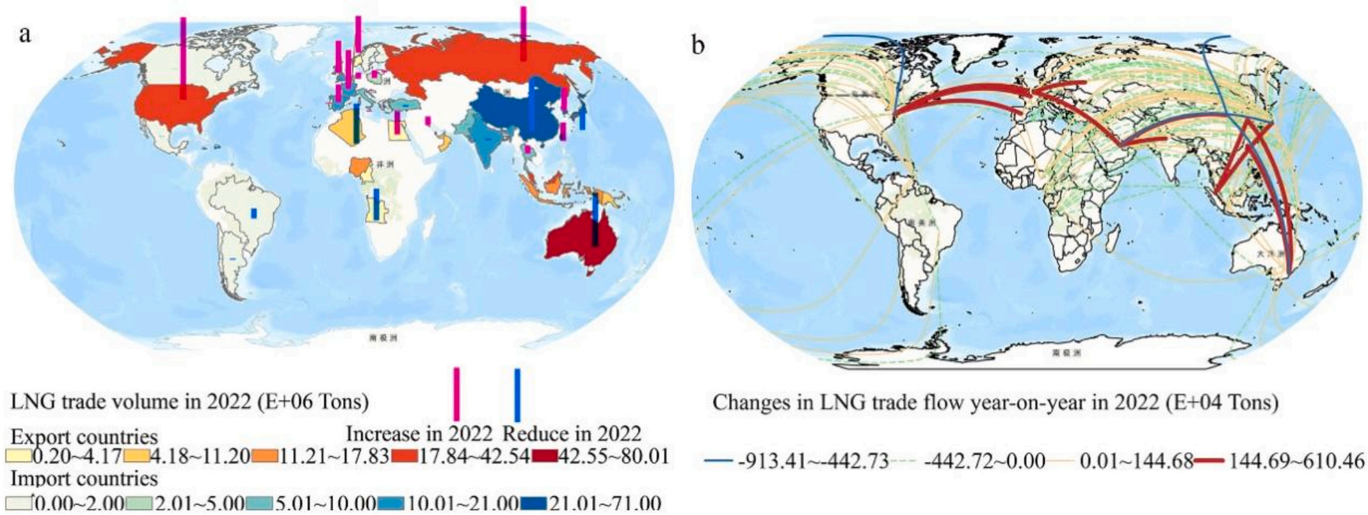


Fig. 2. Changes in Global LNG Trade from 2021 to 2022.

Table 2
Overall indicators of the LNG trade network from 2021 to 2022.

Time	Network density	Average degree	Average weighted degree	Average Path length	Clustering coefficient
2021	0.157	4.113	6,479,259.057	2.335	0.126
2022	0.167	4.426	6,499,386.858	2.219	0.145

vulnerability to sudden events. Policymakers must consider the impact of changes in network structure on national security and economic stability, and seek strategies to strengthen network resilience to external shocks.

The diversification of supply chains augments connectivity within the LNG trade network, thereby enhancing its overall network complexity and interconnectivity. For instance, nations such as France, Spain, and Portugal have broadened their LNG import sources to include supplies from Qatar and United States beyond their conventional channels. The diversification of the supply chain has increased the average degree (average number of connections per node) and network density (ratio of actual edges to maximum possible edges) of the LNG trade network. Additionally, the Russia-Ukraine conflict has coincided with an increase in the average clustering coefficient of the global LNG network, possibly due to increased connectivity within the United States-led "Transatlantic Energy Circle" and the Australian-led "Asian Energy Circle." A detailed analysis of this specific scenario will be provided in Section 4.3.

4.3. *Local structural changes in the LNG trade network*

The Russia–Ukraine conflict has intensified trends within global LNG trade community, thus highlighting the pivotal roles of the United States in the "Transatlantic Energy Circle" and Australia in the "Asian Energy Circle." The conflict has changed the direction and volume of LNG trade, thus leading to a restructuring of the trading communities. Before this conflict, the members of Community 2 were mainly American countries, while the members of Community 3 were mainly Eastern European countries and Russia in 2021 (Fig. 3).

The trend of Russia's isolation in the LNG energy market is gradually emerging. Several European countries were forced to increase their imports of LNG significantly from countries such as the United States (Fig. 2b). Countries such as France, Spain, and Portugal have shifted imports from the former Community 3, where Russia was located, to Community 2, where the United States was located. As a result, the number of members in Community 2 continued to grow, and the number of members in Community 3 decreased in 2022 (Fig. 3).

Notably, the Russia–Ukraine conflict does not exert a significant influence on the structural composition of the Asian LNG trade community. Australia's strategic location between the South Pacific and the Indian Ocean, being closer to Asia than Europe, facilitates its role as a key LNG supplier to the Asian market. Economies such as China, Taiwan (China), Japan, Australia, Malaysia, Indonesia, and others have formed

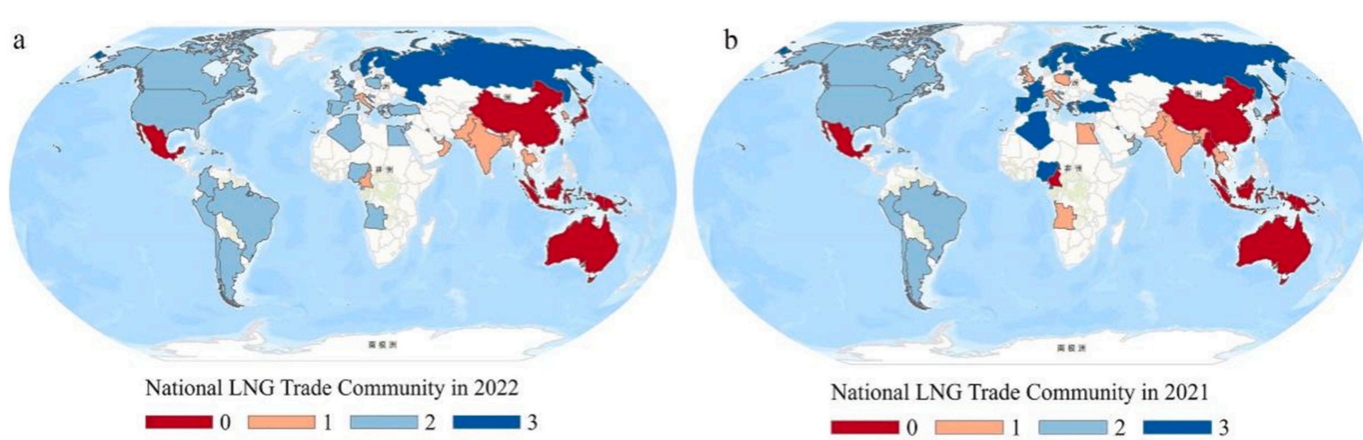


Fig. 3. Changes in the global LNG trade community.

a relatively stable Asian energy circle. In 2022, Russia increased its LNG trade volume with China, while Australia significantly decreased its supply to China (Fig. 2b). Nonetheless, Australia and China continued to be part of the same community predominantly because Australia's LNG import volume to China greatly surpasses that of Russia, with China–Russia trade predominantly occurring through pipeline gas.

5. Factors in the changes of the LNG trade network

The gradual introduction of diverse models is a widely recognized approach in academia, which aids in gaining a deeper understanding of the dynamics within complex networks (Guo et al., 2023a). When investigating the mechanisms of complex networks, scholars (Wang et al., 2024a; Yan et al., 2023) often sequentially incorporate endogenous network effects, exogenous attributes, and external embeddedness networks. This method allows for a more precise revelation of the intricacies involved in the formation and evolution of networks.

Table 3 shows the estimated results of TERGM. Model 1 includes only the endogenous network structure. The endogenous network model focuses on the dynamic processes within the network, without considering external factors. It reveals the internal structure and evolutionary mechanism of the network by analyzing how links in the network form and disappear based on internal factors.

Model 2 adds exogenous country attributes to Model 1. The exogenous country attributes model integrates the characteristics of individual countries within the network into its analysis to investigate how these characteristics influence the establishment of network relationships. The basic principle of this method is that the differences and similarities in individual countries will affect the probability of connections between them, thereby affecting the structure and evolution of the network (Yan et al., 2024).

Model 3 adds the external embeddedness network to Model 2. The external embeddedness network model further considers how relationships in the network are influenced by other network relationships. The basic principle of this method is that the relationships in the network do not exist in isolation, but are embedded and influenced by each other. For example, if two countries already have connections in other networks such as geographical proximity, this may increase the likelihood of them forming a trade relationship (Wang et al., 2024a). The goodness-of-fit test of Tergm shows that the model is suitable (see Appendix B of the Model Test).

We have selected Model 3 as our analytical outcome. Compared to Model 2, Model 3 additionally incorporates the effects of external network embedding, leading to a more comprehensive set of results. Apart from specific variables such as Edges, Gwesp, Nodeicov.NO_2, and Nodeocov.PLE, Model 3 demonstrates consistent trends with Model 2 on the corresponding variables, thereby enhancing the robustness of our study. In the treatment of key variables Edges, Gwesp, and Nodeicov.NO_2, Model 3 yields results that are more closely aligned with real-world conditions. The following sections will provide a detailed analysis.

Table 3
Factors influencing the LNG trade network (*, **, and *** denote significance levels of 10 %, 5 %, and 1 %, respectively, with standard errors of estimated coefficients in parentheses).

	Model 1	Model 2	Model 3
Edges	−0.9656*** (0.0863)	−2.3565*** (0.1733)	3.8047***(0.3719)
Mutual	1.3722**(0.4137)	1.3783**(0.3893)	0.8647**(0.3561)
Gwideg	−1.4122*** (0.0824)	−1.2687*** (0.1258)	−1.3419*** (0.1188)
Gwodeg	−2.2860*** (0.0978)	−1.9433*** (0.1065)	−2.0125*** (0.1179)
Gwdsp	−0.2619*** (0.0442)	−0.2755*** (0.0395)	−0.2837*** (0.0485)
Gwesp	0.4695***(0.0795)	0.1786**(0.0715)	−0.016 (0.0858)
Nodeocov.NO_2		−0.0101** (0.0153)	−0.0496*** (0.0138)
Nodeocov.NL		0.0024***(0.0004)	0.0023***(0.0004)
Nodeocov. PHCA		0.0195***(0.0018)	0.0251***(0.0022)
Nodeocov.PLE		0.0119 (0.0032)	0.0061*(0.0033)
Nodeicov.NO_2		0.0488***(0.0099)	0.0252 (0.0094)
Nodeicov.NL		0.0016**(0.0006)	0.0022***(0.0006)
Nodeicov. PHCA		0.0270***(0.0013)	0.0353***(0.0016)
Nodeicov.PLE		0.0091*(0.0045)	0.0124**(0.0051)
Edgecov.GRN			0.0316*(0.0171)
Edgecov.DIS			−0.7125*** (0.0307)
Edgecov.CON			0.8120***(0.2943)
Edgecov.FTA			0.3100**(0.1246)
Edgecov.GOV			−0.1245*** (0.0174)
Edgecov.CUL			−0.3444** (0.1478)
Stability	2.0323***(0.0385)	1.7791***(0.0336)	1.6435***(0.0320)
Variability	0.0025***(0.0051)	0.0012***(0.0060)	−0.0019 (0.0068)

5.1. Results of endogenous structural effects

The positive outcome of the Edges result indicates that the expansion of edges within the LNG trading network plays a constructive role in the network's formation. In addition, the coefficient of Mutual is significant and positive, thus statistically verifying the systemic mechanism of reciprocity within the global LNG trade network. The concept of reciprocity can be illustrated through the trade relationship between Jamaica and the United States. Jamaica, endowed with abundant LNG reserves, was historically constrained by extraction technology, leading to early imports of LNG from the United States. In recent years, advancements in Jamaica's LNG extraction and transportation technologies have enabled the country to begin exporting LNG to the United States. Jamaican LNG exports play a crucial role in winter energy supply for regions such as Connecticut, Maine, and Massachusetts, especially in the face of pipeline transportation limitations (U.S. Energy Information Administration, 2024). In addition, Japan's Ministry of Economy, Trade and Industry (2024) announced that Japan and South Korea will exchange LNG supplies at certain times, utilizing each other's resources and networks to enhance the stability and security of their LNG provisions. This approach also underscores the notion that reciprocity contributes to the development of supply networks.

The coefficient of Gwdsp is negative and significant, thus indicating that transshipments are not conducive to the development of the LNG trade network. This outcome may be due to the high investment and special storage and transport conditions of LNG, where multiple transshipments significantly increase costs and risks. Furthermore, the insignificance of the Gwesp coefficient suggests that the tendency to form closed triangle trades is not pronounced in the LNG trade network. If countries can trade directly but still need to transit through a third country, this transit mode will increase additional logistics costs and transportation time. This result indirectly confirms the conclusion of the Gwdsp coefficient, which suggests a preference for direct point-to-point trade in the LNG supply chain.

The significant negative coefficients of Gwodeg and Gwideg indicate that the LNG transportation network is highly dependent on a few major countries, which is unfavorable for the network's development. The significant coefficient of Gwodeg indicates an extremely unequal distribution of out-degree centrality in the network, with a few countries supplying most of the world's LNG. Similarly, the significant coefficient of Gwideg indicates that a few countries consume the vast majority of LNG.

The absolute value of the Gwodeg coefficient is greater than that of Gwideg, thus indicating that the negative impact of a few major supplier countries is greater than that of a few major consumer countries. Countries such as the United States have significant bargaining power and influence. The highly asymmetric and overly interdependent structure of this LNG supply chain system causes difficulty in achieving a balanced and efficient global allocation of LNG resources. In the long run, this structure will constrain and threaten the sustained and healthy development of the LNG trade network.

5.2. Results of exogenous national attribute effects

The coefficient of Nodeocov.No_2 and Nodeicov.No_2 indicates that environmental pollution from exporting countries significantly negatively affects the LNG trade network, while pollution from importing countries has no significant effect on the network. For example, the United States was the world's largest LNG exporter in 2023. The United States government announced that it would suspend the approval process for LNG projects to combat climate change and reduce greenhouse gas emissions in 2024 (The White House, 2024). This move will undoubtedly have a direct negative impact on the LNG trade network. LNG-importing countries can mitigate environmental pollution through measures such as reducing energy consumption, promoting energy upgrades, and enhancing environmental governance. Increasing LNG imports is not mandatory in this regard. Therefore, the Nodeicov.No_2 coefficient is not significant.

The coefficients of Nodeocov.PHCA (Nodeicov.PHCA) and Nodeocov.PLE (Nodeicov.PLE) are both significant and positive, with Nodeocov.PHCA (Nodeicov.PHCA) having a greater value than Nodeocov.PLE (Nodeicov.PLE). This indicates that in the development of the LNG trade network, the role of port capacity in promotion is stronger than that of port efficiency. LNG needs to be stored in a low-temperature environment of around minus 160 degrees Celsius, which imposes more complex requirements on port infrastructure. Currently, the global volume of LNG trade continues to grow, while the expansion of port numbers and capacity is struggling to compete with demand. For example, Germany does not have an LNG terminal until 2022.

The coefficients of Nodeocov.NL and Nodeicov.NL are positive and significant but close to zero. This outcome indicates that a country's level of economic development has a limited positive impact on LNG trade. Currently, many less-developed countries still depend on fossil fuels. According to environmental Kuznets curve theory, as countries reach a certain level of economic development, their focus on environmental management increases, thus leading to a gradual transition toward utilizing clean energy.

5.3. Results of external network embedding effect

The positive Edgecov.CON coefficient suggests that geographical proximity enhances LNG trade network development. Adjacency diminishes LNG transport distances and costs while minimizing cultural disparities. Moreover, the significantly negative Edgecov.DIS coefficient suggests that long trading distances impede trade network development, thus indirectly confirming the beneficial impact of geographical proximity in LNG trade.

The positive coefficient of Edgecov.FTA indicates that the participation of trading countries in free trade agreements (FTAs) positively promotes the LNG trade network. Within the framework of FTAs, cross-border capital flows and investments in LNG and related industries encounter fewer barriers, thus increasing the manageability of transportation costs. At the same time, FTAs mitigate the political and legal risks associated with LNG supply, thereby fostering global cooperation on LNG trade and infrastructure.

The positive coefficient of Edgecov.GRN indicates that favorable geopolitical relations among nations play a crucial facilitating role in the development of the global LNG trade network. Generally, stable geopolitical relations between countries contribute to the smooth operation of energy supply chains, thus reducing the potential for geopolitical risks and transportation disruptions. Conversely, deteriorating geopolitical relations may exacerbate the risks of energy supply interruptions or disruptions.

The negative coefficient of Edgecov.GOV suggests that large governance gaps between countries significantly hinder the LNG trade network. Insufficient governance levels can result in instability in transportation and transactions, which consequently raises the risks and costs of international trade. For example, Nigeria, despite having Africa's largest natural gas reserves, faces constraints on its LNG exports because of political instability and weak governance.

The negative implication of the Edgecov.CUL coefficient is consistent with the theory of Hofstede (1994) that significant cultural differences between countries have a negative impact on international business cooperation. Finally, the significantly positive coefficient of network stability and the nonsignificant variability results indicate the stability of the LNG trade network. In other words, the evolution path of the LNG trade network is a continuous and gradual development rather than a sudden leap.

6. Discussion

This study provides new evidence of the changes in the LNG trade network under the influence of geopolitical conflicts. Previous research indicates that the Americas form a largely self-contained LNG trade community, thus achieving regional trade self-sufficiency and integration. However, the Russia–Ukraine conflict has directly altered trade flows and reshaped the network community structure, particularly the integration of European countries into the trade communities of the Americas. Moreover, our results further confirm the positive role of favorable geopolitical relations in the development of trade network, which is consistent with the research findings of Wang et al. (2024a) and indirectly validates the reliability of the conclusions of our study.

Our findings offer a novel theoretical perspective on the complexity of the LNG trade network mechanism. Existing studies (Saslavsky and Shepherd, 2014) have emphasized the critical role of logistics in the production networks. However, this study reveals that port handling capacity ability significantly impacts the global trade network rather than port logistics efficiency. This conclusion holds significant practical importance for the advancement of the LNG trade network. In the process of enhancing the development of global LNG trade, it is imperative for nations to place greater emphasis on increasing the throughput capacity of LNG ports, which is more critical than improving logistics efficiency.

While Guo et al. (2023a) found the impact of national environmental pressures on the LNG trade network to be insignificant, we offer a novel perspective by highlighting previously overlooked asymmetrical effects. Specifically, exporting countries' environmental conditions significantly depress the trade network, which contrasts with the nonsignificant impact of importing countries' environmental conditions. To mitigate the impact of potential reductions in LNG exports from exporting countries due to environmental concerns, importing nations should adopt a strategy of diversifying their supply channels to decrease reliance on a single exporting country. Additionally, importing countries should establish strategic LNG reserves to cope with supply fluctuations and engage in regional cooperation to share resources, thereby facilitating mutual support in emergencies. Furthermore, accelerating the development of clean energy is equally crucial, as it will contribute to long-term energy security and stability.

7. Conclusion

By examining the characteristics and mechanisms of LNG trade network changes in the context of the Russia–Ukraine conflict, this study not only deepens our understanding of LNG trade network dynamics but also underscores its practical significance for global LNG supply security. The Russia–Ukraine conflict has led to a dramatic change in the global LNG trade pattern. We draw the following conclusions. (1) In 2022, the United States shifted its LNG export focus from Asia to neighboring Europe. Many European countries have transitioned from the Russian to the US trade community. The average path length of the network has decreased, thus improving network efficiency. (2) Exogenous national attribute effects, including environmental issues in exporters and port capacities in both countries, notably affect LNG trade network development. Conversely, national GDPs minimally promote trade networks. (3) Endogenous structural effects, such as reciprocity (Mutual), foster the LNG trade network, while traits such as activity (Gwodeg) and connectivity (Gwdsp) can be detrimental. (4) External network embedding effects, such as geopolitical, cultural, and governance factors between countries, have hindered the development of the LNG trade network.

Our study still has some limitations. On the one hand, the relatively short time series of the study because of data availability constraints may introduce certain biases in the results. On the other hand, the LNG trade network is influenced by factors such as natural gas inventories and extreme weather events, which were not thoroughly investigated in this study. For example, in 2022, extreme summer temperatures in Europe led to an increase in LNG consumption in the region. However, abundant natural gas stocks in Europe in the same year effectively alleviated gas supply shortages for winter heating (Xiao et al., 2024). Overall, future research should incorporate factors such as natural gas stocks and extreme weather events into a comprehensive long-term framework to evaluate the mechanisms that influence the LNG trade network.

CRediT authorship contribution statement

Renrong Xiao: Writing – original draft, Methodology, Conceptualization. **Pengjun Zhao:** Writing – review & editing, Supervision, Funding acquisition. **Kangzheng Huang:** Writing – original draft, Conceptualization. **Tianyu Ma:** Writing – original draft, Conceptualization. **Zhangyuan He:** Writing – review & editing. **Caixia Zhang:** Writing – review & editing. **Di Lyu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix A

A.1. AIS data processing

Port logistics efficiency is defined as the time spent by port handling units loading and unloading LNG per unit of weight. This work calculates the port time spent on LNG handling for all vessels in each month based on vessel arrival and departure patterns. Then, the data are aggregated to give the total time spent on LNG loading and unloading for each port in specific countries. Finally, the logistics efficiency indicator for ports in different countries is obtained as follows:

$$PE_{ij} = \frac{\sum PHCA_{ij}}{\sum Parkingtime_{ij}} \tag{5}$$

where $PHCA_{ij}$ represents port handling capacity ability of LNG for country i in month j. $Parkingtime_{ij}$ represents the total parking time for LNG in country i in month j.

A.2. Calculation of nighttime light data

Black Marble nighttime lights are global nighttime imagery products created based on nighttime data from the NPP satellite's VIIRS sensor. The Black Marble nighttime lights product has a spatial resolution of 15 arc *sec*. It provides daily, monthly, and yearly products to meet various research needs at different time scales (Li et al., 2022). Notably, Black Marble nighttime light data are synthesized from multiple repeated observations of the same day by the VIIRS sensor, which effectively reduces the interference from factors such as clouds and aerosols (Iqbal et al., 2022). Therefore, this study has selected the Black Marble nighttime light product for processing. The specific processing steps are as follows:

First, the brightness values of the night lights for each day in 2021 and 2022 are extracted from the Black Marble product VNP46A2 dataset. Second, the total brightness value for each day within the boundaries of each country is extracted based on the administrative boundary grid of each country. Finally, the monthly average nighttime light brightness for each country is calculated, which is used as an indicator variable to measure the level of economic development in the country.

A.3. Calculation of NO_2 concentration

TROPOMI, which was installed on the Copernicus satellite launched in 2017, is an atmospheric monitoring instrument that measures atmospheric components and greenhouse gases. The spatial resolution of TROPOMI products is 7 by 7 km with a spectral resolution of 0.25–0.50 nm (Butz et al., 2012). TROPOMI covers ultraviolet, visible, near-infrared, and shortwave infrared spectral bands, thus enabling the monitoring of gases such as CH_4, CO, O_3, NO_2, SO_2, and HCHO (Veefkind et al., 2012). Therefore, TROPOMI products are selected for processing in this study. The specific processing steps are as follows:

First, daily TROPOMI remote sensing data products for 2021 and 2022 are obtained from the Sentinel-5P satellite. Second, daily NO_2 concentrations are extracted for the country using national administrative boundaries. Finally, the monthly average NO_2 concentration for each country is calculated as an indicator variable to measure the level of atmospheric pollution in each country.

Appendix B

B.1. Model test

This study conducted a goodness-of-fit test on the results of the Tergm model. The geodesic distance, dyad-wise shared partners, edge-wise shared partners, indegree, and triad census are selected as key network features. Based on Model 3, 1000 simulations are run for each month's LNG trade network, followed by a comparison of the simulated network characteristics with those of the observed network.

The black line in Fig. 4 represents the calculated results of the observed network, while the grey lines represent the calculated results for the simulated network within the 95 % confidence interval. The black line generally falls within the box, thus indicating that the simulated network adequately reflects the structural characteristics of the observed network. Receiver operating characteristic (ROC) and precision-recall curves are plotted in this study to assess the goodness-of-fit of Tergm. The red curve in the graph represents the ROC curve, while the blue curve represents the PR curve. The general assumption is that the closer the ROC curve is to the upper left corner of the graph, the better the fit of the model is. The results in Fig. 3 clearly demonstrate the good fit of Model 3.

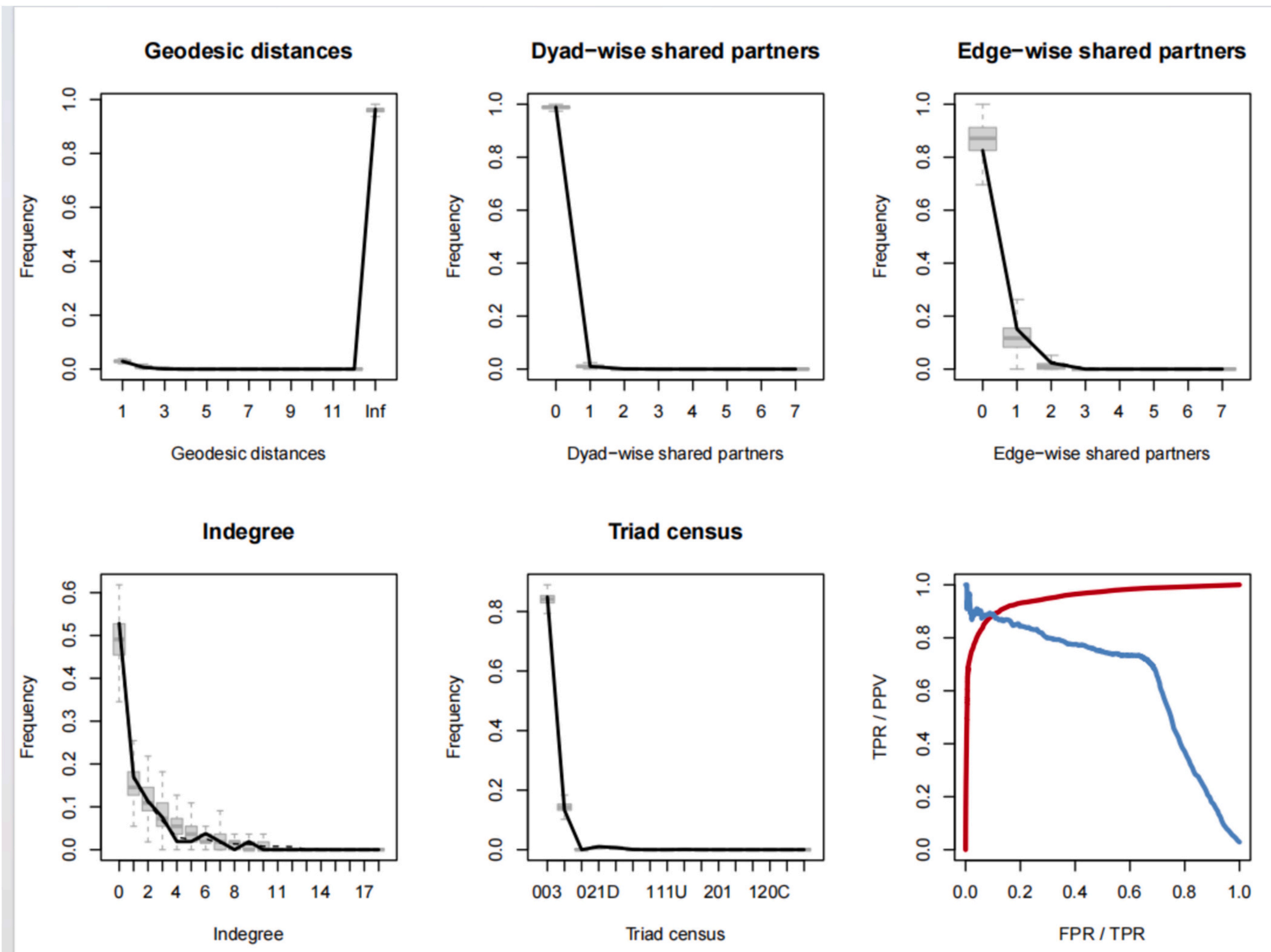

Fig. 4. Goodness-of-fit diagnostics for Tergm.

In this study, we apply the maximum pseudo-likelihood estimation (MPLE) to conduct robustness tests on the results of the Tergm model. Given that the samples for empirical analysis include more than 50 countries over 20 months, using the MPLE is preferred when dealing with a big sample (Wang et al., 2024a). Specifically, Model 4 is tested using data from January to December 2021, Model 5 is tested using data from January to February 2022, and Model 6 is tested using data with a two-month interval from 2021 to 2022 (Table 4). Models 4 to 6 consistently demonstrate the high robustness of the conclusions presented in this study.

Table 4
Robustness detection results of the TERGM model (*, **, and *** denote significance levels of 10 %, 5 %, and 1 %, respectively, with standard errors of estimated coefficients in parentheses).

Classification	Variables	Model 4	Model 5	Model 6
Endogenous Structural Statistics	Edges	3.7604***(0.7375)	3.1250***(0.4546)	2.7969***(0.6308)
	mutual	0.5806 (0.5752)	1.0914**(0.4127)	1.0460*(0.5739)
	Gwideg	−1.2999***(0.1792)	−0.8530***(0.2465)	−1.0115***(0.1700)
	Gwodeg	−2.3044***(0.1874)	−1.4923***(0.1785)	−1.7410***(0.1750)
	Gwdsp	−0.2587***(0.0612)	−0.3466***(0.0547)	−0.2751***(0.0675)
	Gwesp	−0.0258 (0.1325)	−0.0927 (0.0786)	−0.1183 (0.1280)
Exogenous national attribute effects	Nodeocov.NO_2	−0.0217 (0.0301)	−0.0587***(0.0141)	−0.0415**(0.0216)
	Nodeocov.NL	0.0025***(0.0005)	0.0019***(0.0004)	0.0025***(0.0003)
	Nodeocov.PHCA	0.0306***(0.0018)	0.0267***(0.0037)	0.0291***(0.0019)
	Nodeocov.PE	0.0000 (0.4472)	1.2672***(0.1649)	0.9505**(0.3112)
	Nodeicov.NO_2	0.0230 (0.0192)	0.0002 (0.0182)	0.0157 (0.0166)
	Nodeicov.NL	0.0039***(0.0009)	0.0017*(0.0010)	0.0028**(0.0010)
	Nodeicov.PHCA	0.035***(0.0032)	0.0376***(0.0020)	0.0339***(0.0027)
	Nodeicov.PE	0.0000 (0.5828)	1.3055***(0.1963)	1.1983***(0.2497)
External network embedding effect	Edgecov.GRN	−0.0025 (0.0299)	0.0575**(0.0199)	0.0343**(0.0151)
	Edgecov.DIS	−0.7282***(0.0547)	−0.8210***(0.0365)	−0.7698***(0.0496)
	Edgecov.CON	1.2839***(0.4003)	0.3906 (0.3669)	0.8528**(0.4291)
	Edgecov.FTA	0.2161 (0.1840)	0.2027 (0.1497)	0.1721 (0.1745)
	Edgecov.GOV	−0.1608***(0.0280)	−0.0937**(0.0413)	−0.1325***(0.0288)
	Edgecov.CUL	−0.5858**(0.2354)	−0.1640 (0.1170)	−0.3392 (0.2456)
Time effect	Stability	1.7021***(0.0469)	1.5643***(0.0307)	1.5921***(0.0552)
	Variability	0.0023 (0.0237)	−0.0086 (0.0228)	−0.0156 (0.0137)

Data availability

The authors do not have permission to share data.

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